Prelab 5

9.1

$y^{(3)} = 3\ddot{y} + 3\dot{y} + y = 5u$

$x_1 = y \quad \dot{x}_1 = \dot{y} = x_2 \qquad \dot{x}_2 = \ddot{y} = x_3 \qquad \dot{x}_3 = y^{(3)} = -3x_3 - 3x_2 - x_1 + 5u$

$\dot{x}_3 = y^{(3)} = -3\ddot{y} - 3\dot{y} - y + 5u$

$= -3x_3 - 3x_2 - x_1 + 5u$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -3 & -3 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

$$C = [1 \quad 0 \quad 0]$$

9.2

$\ddot{f} + 2\dot{f} + 3f + 3g = u_1 + u_2$

$\dot{g} + 3\dot{f} + g = u_2$

$x_1 = f$

$x_2 = \dot{f}$

$x_3 = g$

$\dot{f} = u_1 + u_2 - 2\dot{f} - 3f - 3g$

$\dot{x}_2 = u_1 + u_2 - 2x_2 - 3x_1 - 3x_3$

$\dot{g} = -3\dot{f} - g + u_2$

$\dot{x}_3 = -3x_2 - x_3 + u_2$

$\dot{x}_1 = \dot{x}_2 \qquad \dot{x}_2 = -3x_1 - 2x_2 - 3x_3 + u_1 + u_2$

$\dot{x}_3 = -3x_2 - x_3 + u_2$

5.2 controllable

$$A = \begin{bmatrix} 0 & 1 & 0 \\ -3 & -2 & -3 \\ 0 & -3 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

5.3 $\dot{x} = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix} x + \begin{bmatrix} 1 \\ -2 \end{bmatrix} u$

$$y = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x$$

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} + \begin{bmatrix} -1 & 2 \\ -3 & -4 \end{bmatrix} = \begin{bmatrix} s-1 & 2 \\ -3 & s-4 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{\Delta} \begin{bmatrix} s-4 & -2 \\ 3 & s-1 \end{bmatrix}$$

$$\Delta = (s-1)(s-4) - (2)(-3) = s^2 - 5s + 10$$

$$H(s) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} s-4 & -2 \\ 3 & s-1 \end{bmatrix} \cdot \frac{1}{s^2 - 5s + 10} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{s-5}{s^2 - 5s + 10} \\ \frac{s-3}{s^2 - 5s + 10} \end{bmatrix}$$, work on next page

Prof

3.3 continued

$$= \begin{bmatrix} s-4 & 3 \\ 0 & s-1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} \cdot \frac{1}{s^2-5s+4}$$

$$[I_s - A]^{-1} \cdot B = \frac{1}{\Delta} \begin{bmatrix} s-1 & -2 \\ 3 & s-1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} s-4+4 \\ 3-2s+2 \end{bmatrix}$$

$$= \frac{1}{s^2-5s+10} \cdot \begin{bmatrix} s \\ 5-2s \end{bmatrix}$$

$$C x^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} s \\ 5-2s \end{bmatrix} \cdot \frac{1}{\Delta} = \begin{bmatrix} s+5-2s \\ 5-2s \end{bmatrix} \cdot \frac{1}{\Delta}$$

$$= \begin{bmatrix} \frac{1}{s^2-5s+10} & \begin{bmatrix} 5-s \\ 5-2s \end{bmatrix} \end{bmatrix}$$

5.4

$$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s-1} & 0 & 0 \\ 0 & \frac{1}{s+2} & 0 \\ 0 & 0 & \frac{1}{s+3} \end{bmatrix}$$

$$H(s) = \begin{bmatrix} 1 & 3 & -4 \end{bmatrix} \begin{bmatrix} \frac{1}{s-1} & 0 & 0 \\ 0 & \frac{1}{s+2} & 0 \\ 0 & 0 & \frac{1}{s+3} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$= C \cdot (sI - A)^{-1} \cdot B = \begin{bmatrix} \frac{1}{s-1} & \frac{3}{s+2} & \frac{5}{s+3} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$= \frac{1}{s-1} + \frac{9}{s+2} + \frac{-20}{s+3} = H$$

5.5 $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$

$$sI - A = \begin{bmatrix} s-1 & 2 \\ -3 & s-4 \end{bmatrix}$$

$$\det(sI - A) = (s-1)(s-4) - (-6)$$

$$= s^2 - 5s + 10 = 0$$

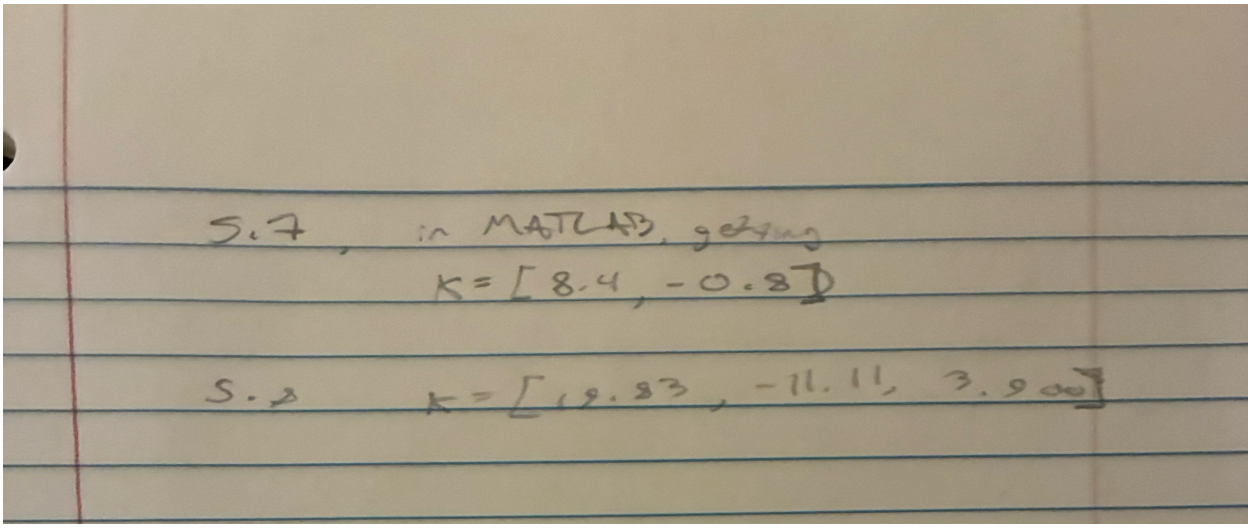
$$s = \frac{5 \pm \sqrt{25 - 40}}{2}$$

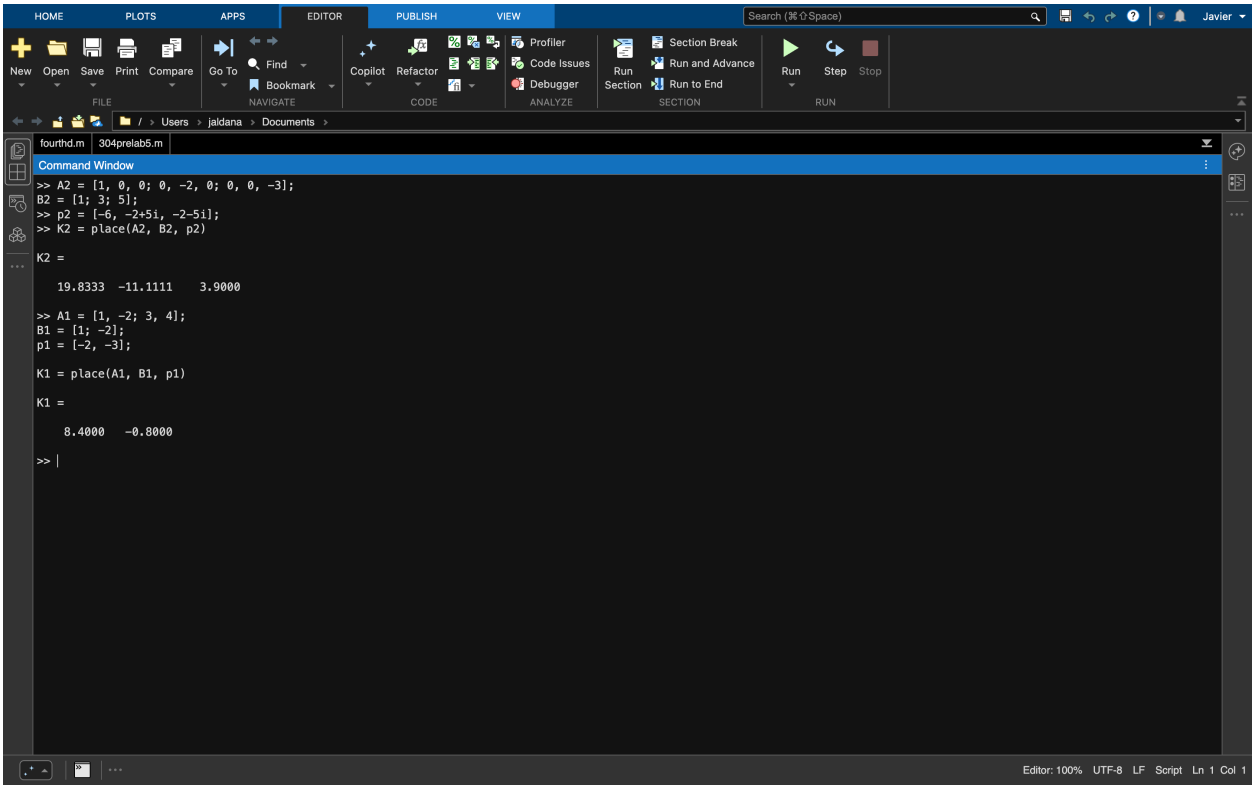
5.6 $sI - A = \begin{bmatrix} s-1 & 0 & 0 \\ 0 & s+2 & 0 \\ 0 & 0 & s+3 \end{bmatrix}$

$$\Rightarrow \det(sI - A) = (s-1)(s+2)(s+3) = 0$$

$s_1, s_2, s_3 = 1, -2, -3$

The poles make $H(s)$ ∞ @ some value


5.7 , in MATLAB, getting
K = [8.4, -0.8]
5.8 K = [19.83, -11.11, 3.900]


HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
Search (⌘⇧Space)
Javier
New
Open
Save
Print
Compare
Go To
Find
Bookmark
Copilot
Refactor
Profiler
Code Issues
Debugger
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
SECTION
RUN
/ › Users › jaldana › Documents ›
fourthd.m
304prelab5.m
Command Window
>> A2 = [1, 0, 0; 0, -2, 0; 0, 0, -3];
B2 = [1; 3; 5];
>> p2 = [-6, -2+5i, -2-5i];
>> K2 = place(A2, B2, p2)
K2 =
19.8333 -11.1111 3.9000
>> A1 = [1, -2; 3, 4];
B1 = [1; -2];
p1 = [-2, -3];
K1 = place(A1, B1, p1)
K1 =
8.4000 -0.8000
>>
Editor: 100% UTF-8 LF Script Ln 1 Col 1